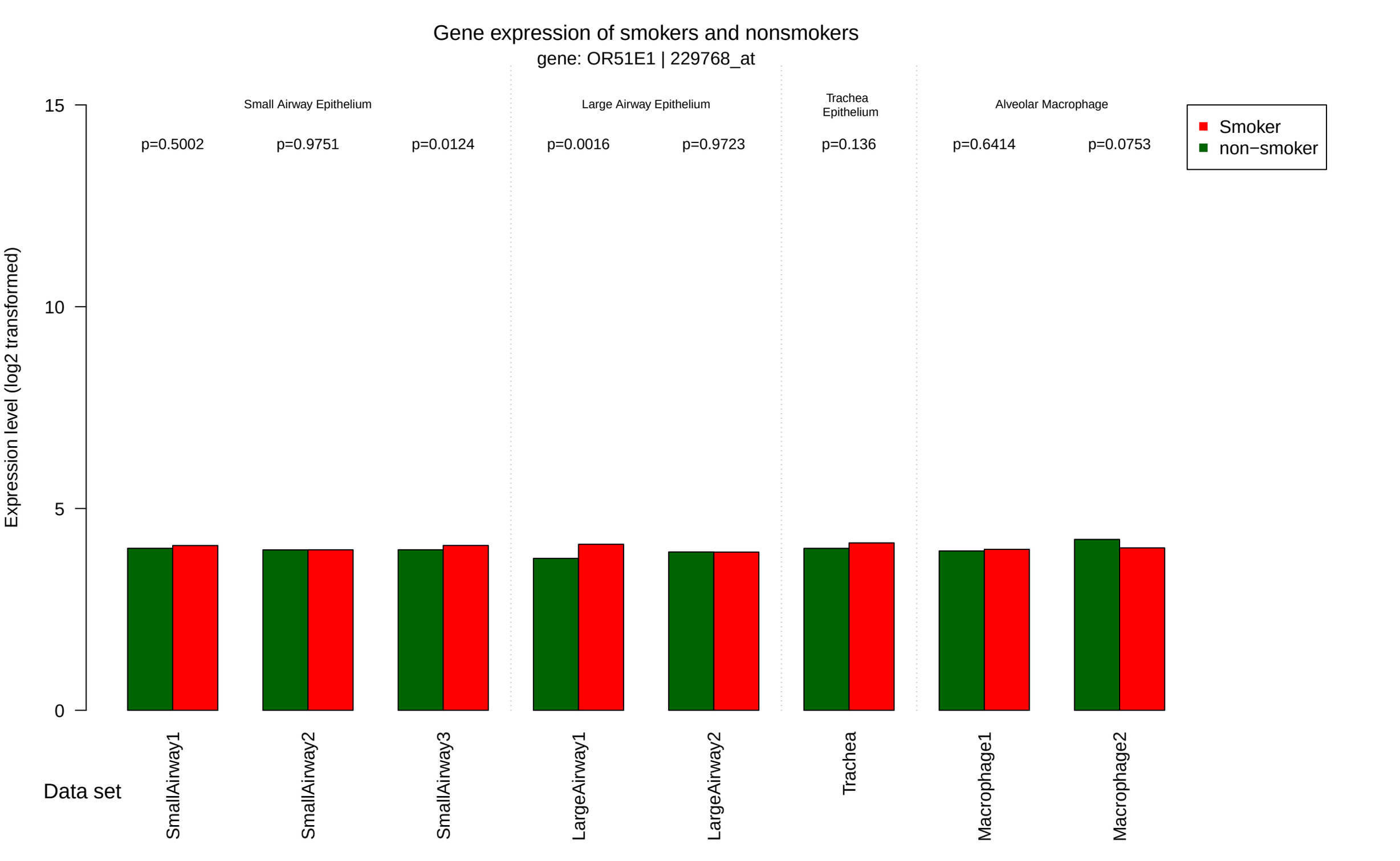
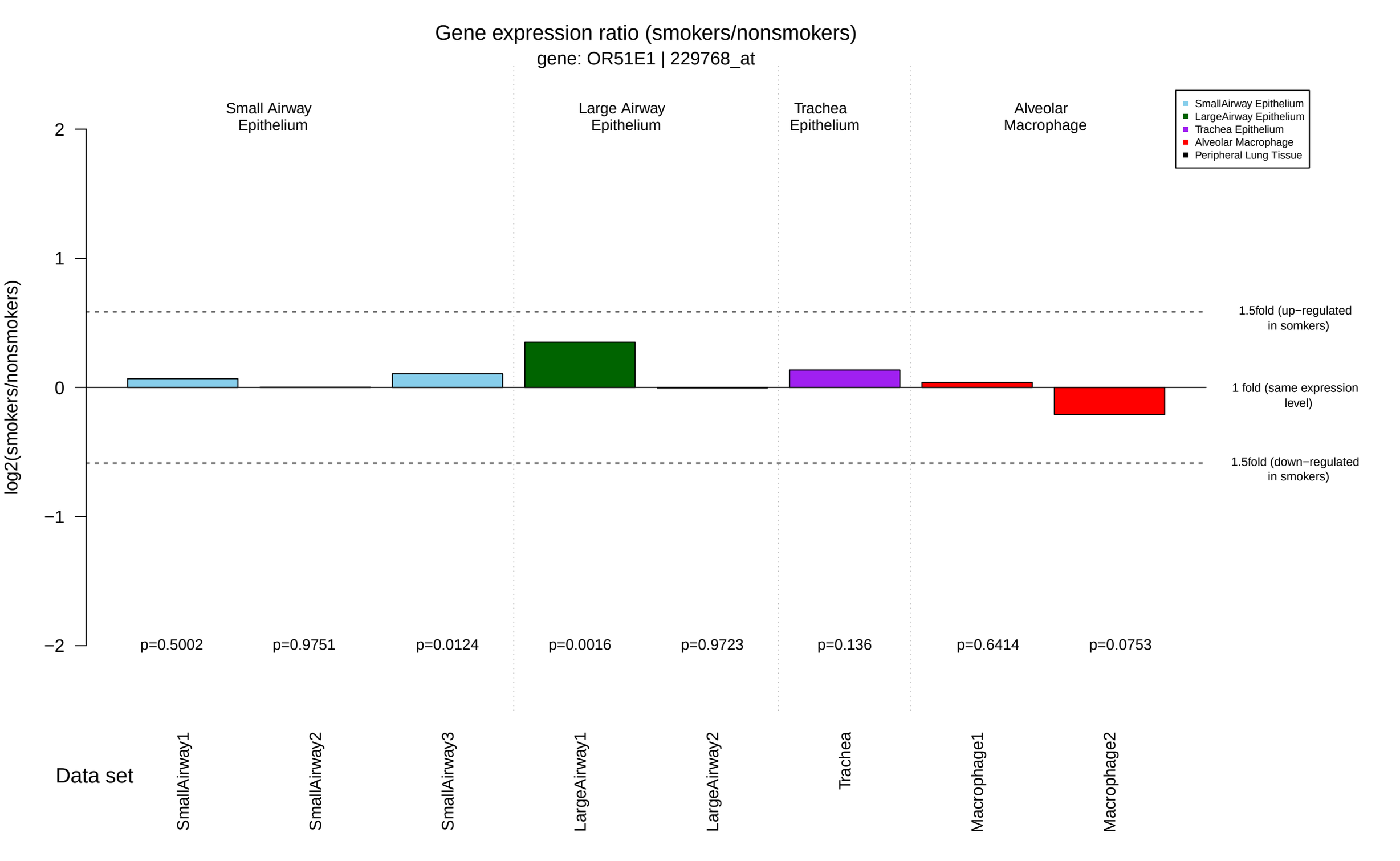
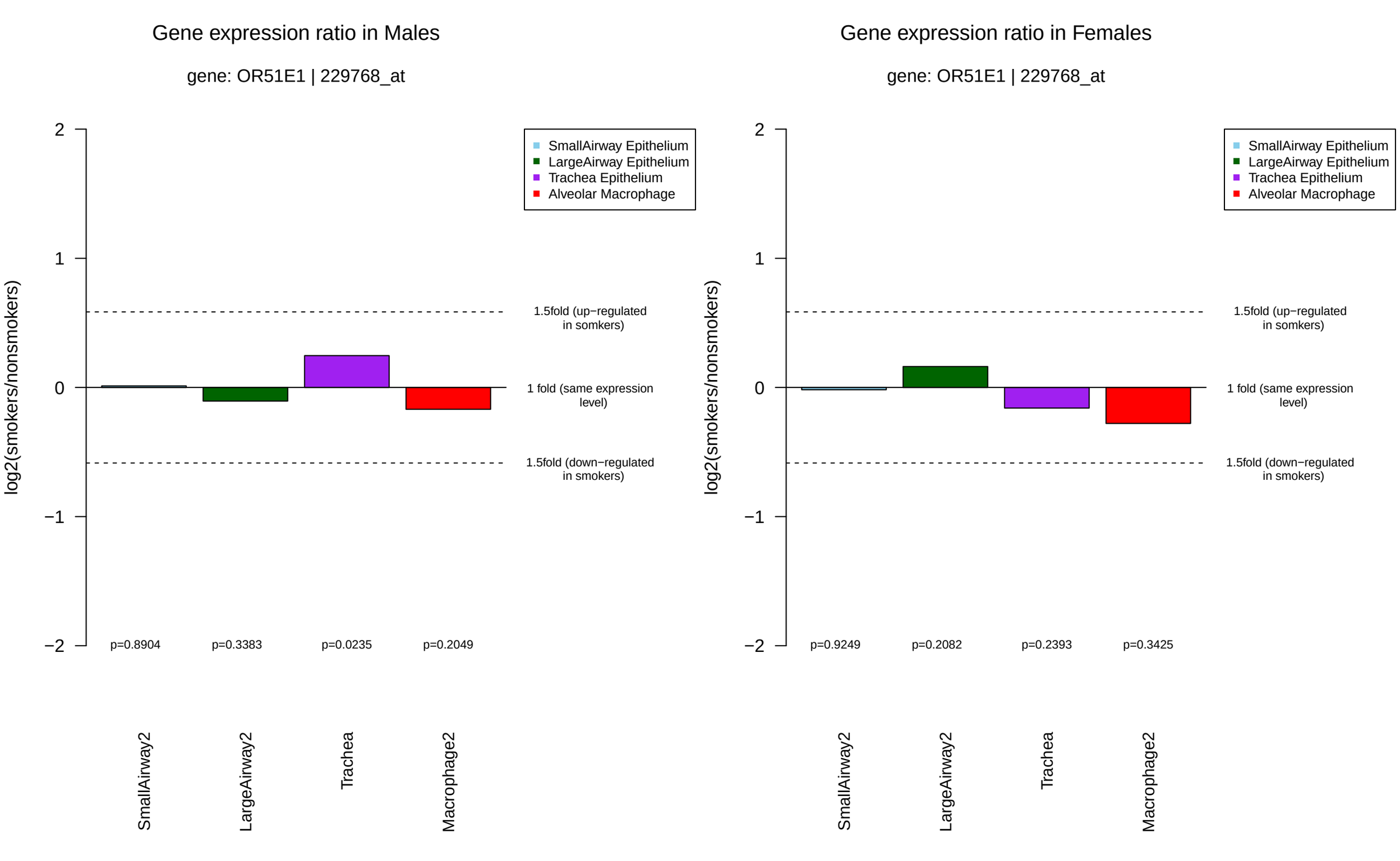
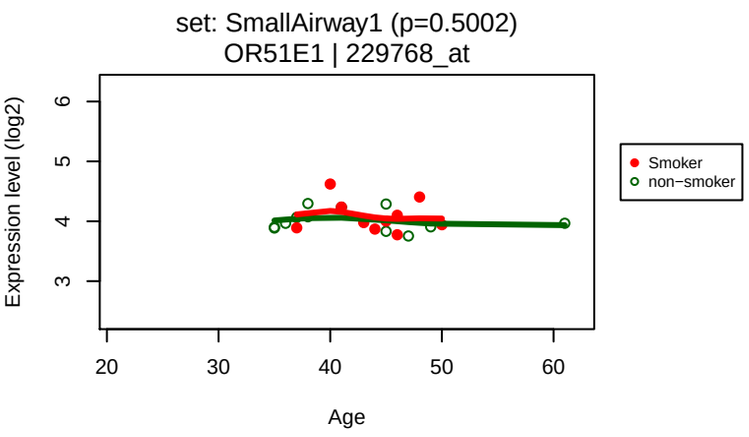
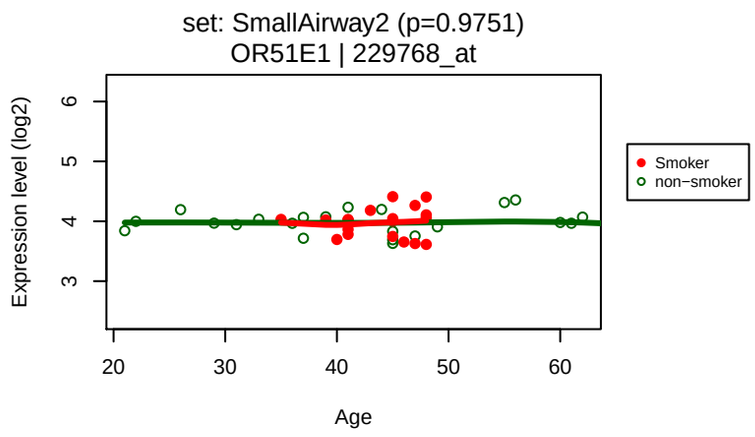
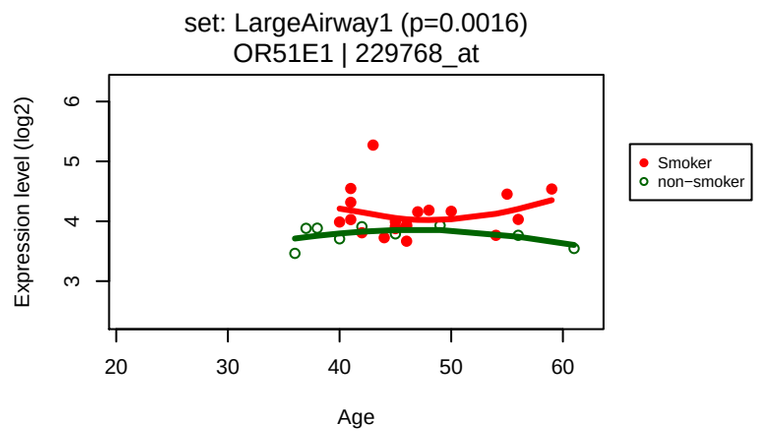

Gene expression of smokers and nonsmokers
gene: OR51E1 | 229768_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5002
p=0.9751
p=0.0124
p=0.0016
p=0.9723
p=0.136
p=0.6414
p=0.0753
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

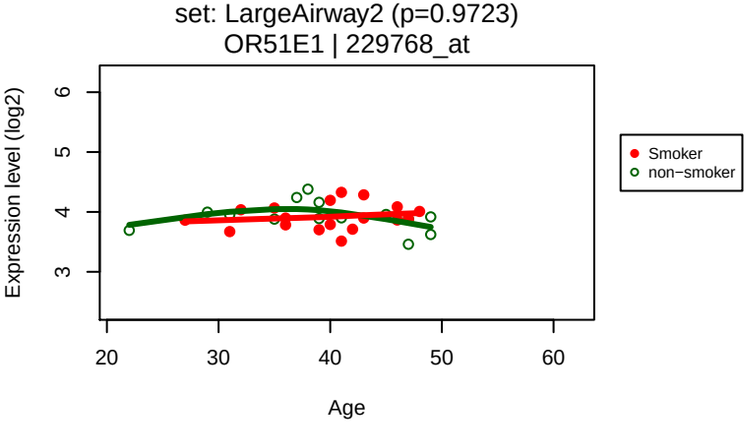

Gene expression ratio (smokers/nonsmokers)
gene: OR51E1 | 229768_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5002
p=0.9751
p=0.0124
p=0.0016
p=0.9723
p=0.136
p=0.6414
p=0.0753
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

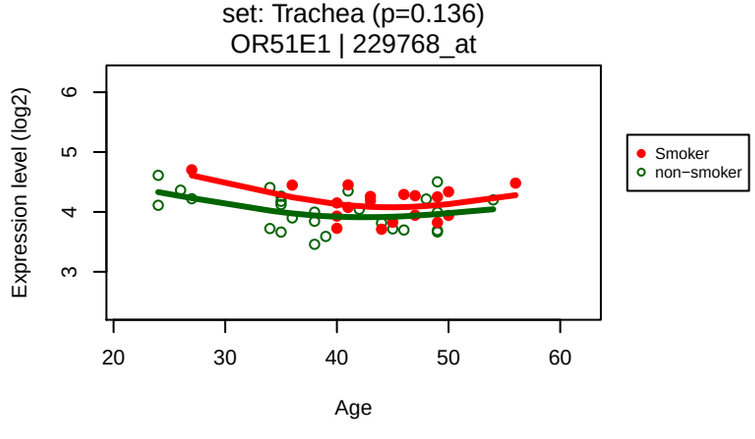

Gene expression ratio in Males
gene: OR51E1 | 229768_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8904
p=0.3383
p=0.0235
p=0.2049
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51E1 | 229768_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9249
p=0.2082
p=0.2393
p=0.3425
SmallAirway2
LargeAirway2
Trachea
Macrophage2

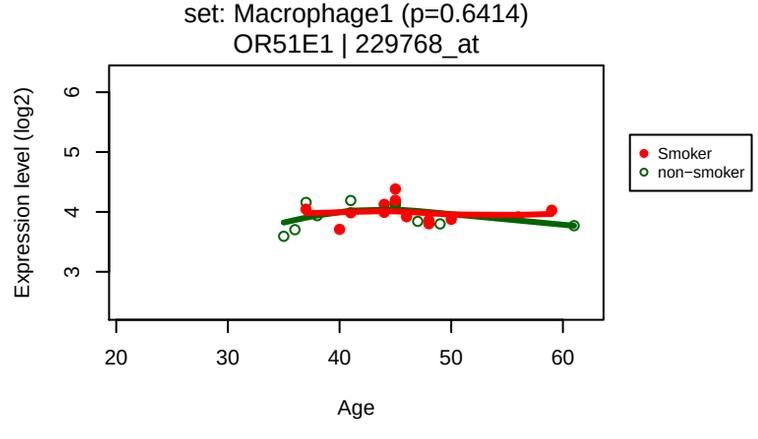

set: SmallAirway1 (p=0.5002)
OR51E1 | 229768_at
Expression level (log2)
Age
Smoker
non−smoker

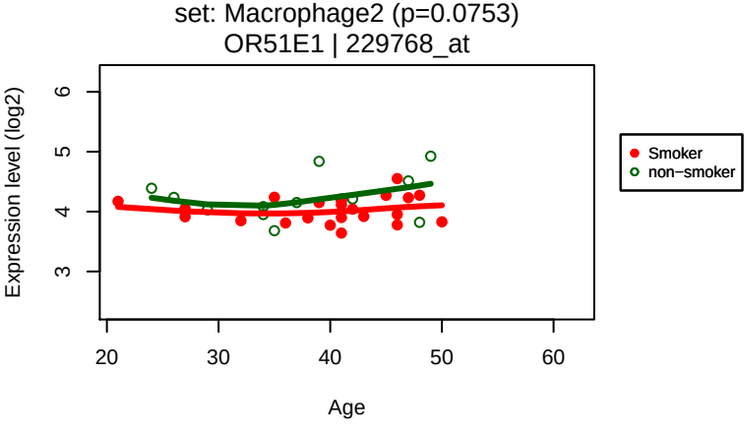

set: SmallAirway2 (p=0.9751)
OR51E1 | 229768_at
Expression level (log2)
Age
Smoker
non−smoker

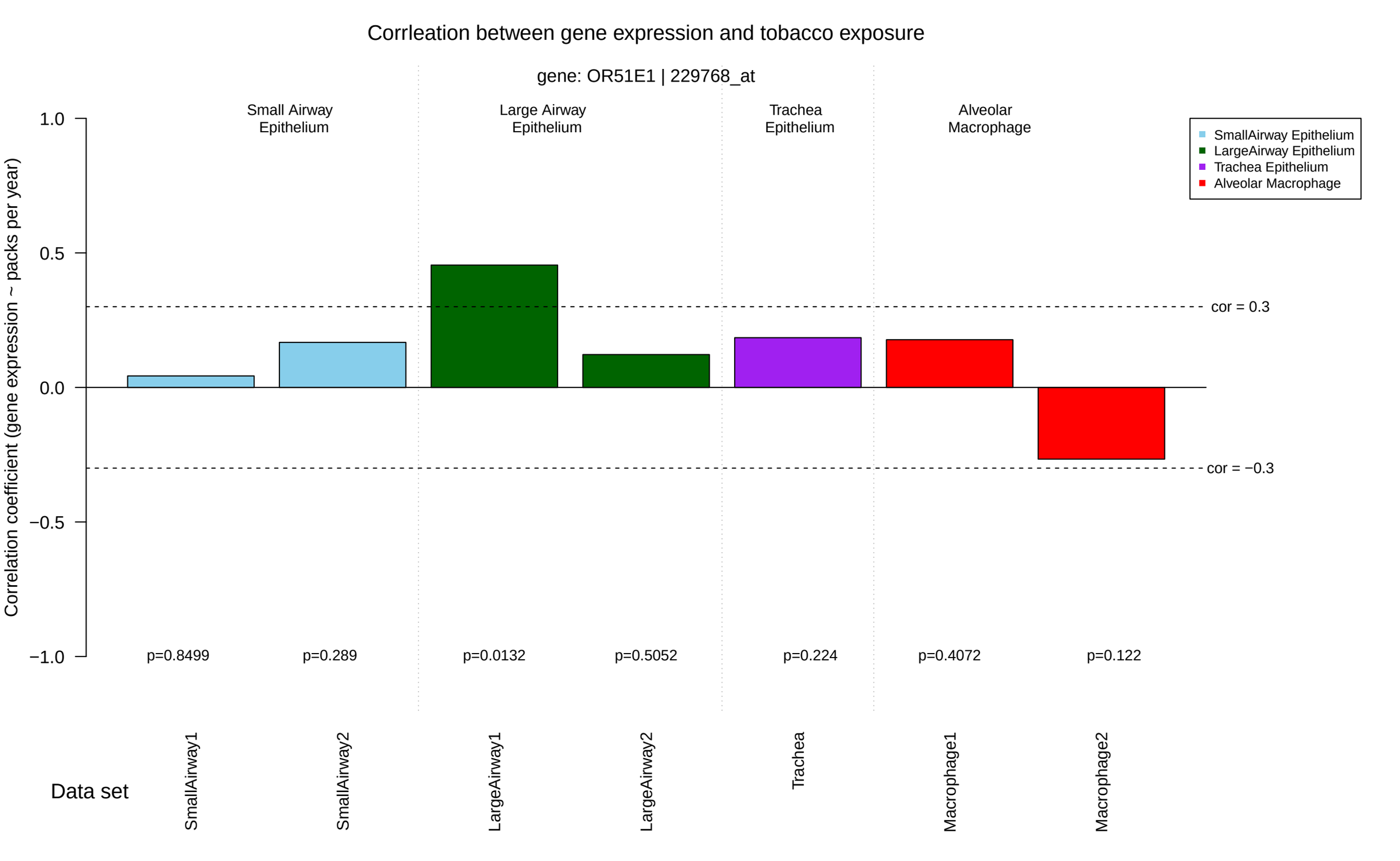

set: LargeAirway1 (p=0.0016)
OR51E1 | 229768_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9723)
OR51E1 | 229768_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.136)
OR51E1 | 229768_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6414)
OR51E1 | 229768_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0753)
OR51E1 | 229768_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51E1 | 229768_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8499
p=0.289
p=0.0132
p=0.5052
p=0.224
p=0.4072
p=0.122
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2